

COSIMA Cookbook: a community resource for ocean and sea-ice modelling

Navid C. Constantinou^{1,4,13*}, Julia Neme^{2*}, Wilton Aguiar², Matthis Auger¹⁶, Ashley J. Barnes^{4,10}, Romain Beucher³, Dhruv Bhagtani¹¹, Christopher Yit Sen Bull³, Hannah Dawson¹⁶, Noah Day¹³, Fabio Boeira Dias¹⁴, Edward W. Doddridge¹⁶, Anupiya Ellepola², Denisse Fierro-Arcos¹⁶, Angus Gibson², Maurice F. Huguenin⁸, Aidan Heerdegen³, Andy McC. Hogg^{2,3,4}, Ryan M. Holmes⁵, Wilma Huneke^{2,10}, Jemma Jeffree^{2,4}, Andrew E. Kiss², Minghang Li³, Josué Martínez-Moreno¹², Jan Jaap Meijer¹⁶, Thomas Moore⁷, Ruth Moorman⁶, Paige Martin³, Adele K. Morrison², James Munroe¹, Aditya Narayanan¹⁵, Micael Oliveira³, Ellie Q. Y. Ong^{4,10}, Madelaine G. Rosevear^{4,16}, Christina Schmidt¹², Taimoor Sohail¹³, Paul Spence^{4,16}, Dougal T. Squire³, Anton Stekete³, Charles Turner³, Felipe Vilela da Silva⁹, Marc White^{2,3}, Luwei Yang², Claire Yung², and Jan Zika¹⁴

¹ Earthdata Associates, Canada ² Australian National University, Australia ³ Australian National University, Australia's Climate Simulator (ACCESS-NRI), Australia ⁴ Australian Research Council Centre of Excellence for the Weather of the 21st Century, Australia ⁵ Bureau of Meteorology, Australia ⁶ California Institute of Technology, USA ⁷ Commonwealth Scientific and Industrial Research Organisation, Australia ⁸ GEOMAR Helmholtz Centre for Ocean Research Kiel, Germany ⁹ IFREMER - French Institute for Ocean Science, France ¹⁰ Monash University, Australia ¹¹ Princeton University, USA ¹² British Antarctic Survey, UK ¹³ University of Melbourne, Australia ¹⁴ University of New South Wales Sydney, Australia ¹⁵ University of Southampton, UK ¹⁶ University of Tasmania, Australia * These authors contributed equally.

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Software

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Summary

The COSIMA Cookbook is an open computational learning module for analysing ocean and sea-ice model output in Jupyter notebooks (Kluyver et al., 2016). It has been developed by the Consortium for Ocean–Sea Ice Modelling in Australia (COSIMA; <https://cosima.org.au>) as a community resource for researchers, students, and practitioners working with large gridded datasets, especially output from the ACCESS ocean–sea ice model configurations (e.g. Kiss et al., 2020). The repository combines introductory tutorials with worked analysis examples, all exposed through a browsable documentation site built with Sphinx, a documentation generator for Python projects (Komiya et al., 2025) (Figure 1) and backed by a citable archived release (Constantinou et al., 2026).

COSIMA Cookbook

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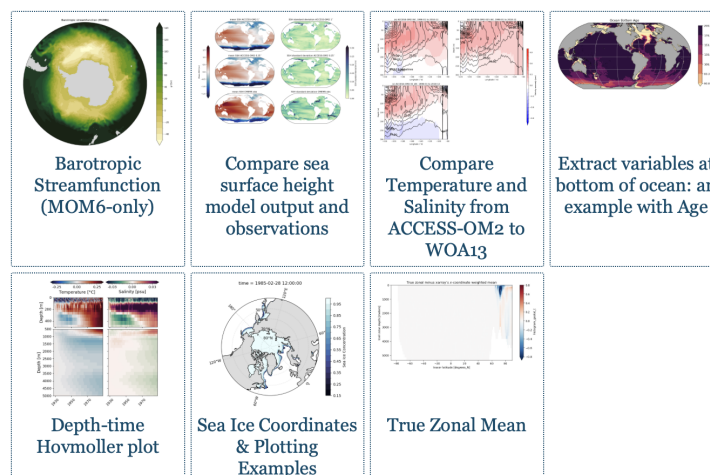
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latest

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Figure 1: The COSIMA Cookbook documentation website, showing the browsable Sphinx-based landing page used to navigate tutorials and recipes. The live site is available at <https://cosima-recipes.readthedocs.io>.

Following the “Cookbook” concept, the website sections are deliberately named using a gastronomy theme. “Cooking Tutorials” refers to tutorials that teach generic, transferable techniques for handling ocean–sea ice model output. Then comes the main part of any cookbook – its recipes. Here, by “recipes” we mean self-contained notebooks showing how to perform concrete diagnostics and analyses on ocean and sea-ice datasets. First come the “Easy Recipes” that provide a good entry point after the tutorials; then the “Advanced Recipes” that are more elaborate and demonstrate advanced analysis techniques, and lastly the “Regional Specialties” contains recipes for regional model configurations (Barnes et al., 2024). Together these sections present the documentation as a browsable collection of lessons and recipes gathered into a single cookbook.

Pedagogy and instructional structure are central to the project. The recipes provide the starting point for more elaborate analyses. “Cooking Tutorials” introduces generic skills such as loading model output, working with labelled arrays, plotting, and interacting with shared data catalogues. These tutorials lead into domain-focused recipes giving learners an incremental path from first contact with the data ecosystem to adaptation of full workflows for their own science questions.

Within this structure, the tutorials leverage and demonstrate open source tools for scientific analysis of large data. This includes examples of loading model output through Intake-based catalogues (Intake developers, 2026), using xarray (Hoyer & Hamman, 2017) and Dask (Dask Development Team, 2016; Rocklin, 2015) to process very large multi-dimensional datasets in parallel. The recipe collection includes a Lagrangian particle-tracking workflow

55 built with Parcels (Delandmeter & Seville, 2019), as well as analyses that use xgcm
56 (Abernathy et al., 2022) to handle variables defined on staggered finite-volume grids,
57 and xESMF (Zhuang et al., 2025) for regridding model output. Figures are generated
58 using matplotlib (Hunter, 2007) incorporating Cartopy (Elson et al., 2024) for map-based
59 visualisation.

60 The Cookbook grew from a practical need inside the COSIMA community: the tools
61 used to analyse modern ocean model output are powerful, but the gap between package-
62 level documentation and reproducible end-to-end workflows remains large. Users need to
63 understand not only Python and Jupyter (Kluyver et al., 2016), but also how to navigate
64 high-dimensional model output, shared high-performance computing environments, and
65 domain-specific analysis conventions. The Cookbook addresses this gap by packaging
66 reusable workflows in the same medium in which users actually work.

67 Statement of Need

68 Ocean and climate model analysis has a steep entry cost. New users must learn how
69 to find datasets, load them efficiently, interpret metadata, operate on multi-dimensional
70 arrays, and produce scientifically meaningful diagnostics. General-purpose libraries such
71 as xarray (Hoyer & Hamman, 2017) provide the foundations for these tasks, but they
72 do not by themselves show a learner how to translate a research question into a robust
73 analysis workflow for a specific model and computing environment.

74 That gap is not only pedagogical; it directly affects how quickly model output can be turned
75 into science. For a beginner, becoming productive with a model often requires substantial
76 time just to understand the structure of the output, the relevant conventions, and the
77 computational environment in which analysis is expected to run. There is also substantial
78 scientific labour embedded in many common diagnostics. Some calculations, for example
79 computing cross-contour transports, require non-trivial development, understanding of
80 model numerics, and validation before they can be used confidently. If every new user or
81 project had to reconstruct those workflows independently, a large amount of effort would
82 be repeatedly spent on rebuilding analysis code rather than conducting research using the
83 model output.

84 The COSIMA Cookbook facilitates knowledge sharing and accelerates research with a
85 domain-specific, openly maintained collection of computational lessons and examples. Its
86 contribution is not a new analysis library; rather, it is a reusable scientific and educational
87 resource that exposes complete, well-documented workflows in the same notebook format
88 in which many researchers actually explore data and communicate methods. By choosing
89 notebooks instead of packaging these examples only as a software library, the project
90 makes intermediate reasoning, methodological choices, and practical execution details
91 visible to learners while still giving more experienced users analysis patterns they can
92 adapt directly for research. This aligns well with JOSE's emphasis on open educational
93 resources that enable computational learning through authentic practice.

94 Several design decisions make the Cookbook particularly useful for adoption by other groups.
95 First, each notebook is intended to be self-contained and well-documented, so learners
96 can read, run, and modify a complete workflow without reconstructing missing context
97 from scattered notes. Second, the repository distinguishes tutorials from recipes. Tutorials
98 teach transferable skills, while recipes demonstrate how those skills combine in realistic
99 analysis tasks. Third, the project includes contributor guidance and notebook review
100 conventions that encourage new analyses to be written in a reusable and pedagogically
101 clear style. A caveat is that some recipes use high-resolution model output with datasets
102 of order terabytes, so the underlying data are not always practical to distribute or mirror
103 in full. However, beyond the initial data-loading step, which usually occurs early in each
104 recipe, the analysis workflows are generally applicable to most operational ocean-sea ice

105 model outputs that can be loaded with xarray ([Hoyer & Hamman, 2017](#)).

106 Thus, the structure is valuable beyond the immediate COSIMA context. Many research
107 communities maintain model output on shared infrastructure and face the same challenge:
108 turning expert tacit knowledge into examples that newcomers can adapt. The Cookbook
109 offers a model for how a scientific collaboration can capture that knowledge in version-
110 controlled notebooks, publish it as a living resource, and continuously improve it through
111 community contribution.

112 Educational Design and Experience of Use

113 The learning experience is organised as a progression. The documentation points new users
114 first to introductory lessons, including material on loading, slicing, and visualising model
115 output. Learners then move to more advanced tutorials and finally to recipe notebooks
116 that address concrete scientific questions. The categories of “easy” and “advanced” recipes
117 provide a lightweight pedagogical cue about expected complexity and scope, with “regional
118 specialties” specifically covering recipes for regional model configurations ([Barnes et al.,
119 2024](#)).

120 The intended mode of use is also explicit. The repository is designed around Jupyter-
121 based analysis on the Australian National Computational Infrastructure, with guidance
122 on running notebooks through ARE/JupyterLab sessions and on accessing shared data
123 holdings. This makes the Cookbook more than a static collection of examples: it is
124 operational documentation for a real analysis environment. At the same time, the
125 notebooks demonstrate broadly transferable patterns for working with labelled geophysical
126 data, so individual lessons can be reused or adapted outside that environment.

127 The project also supports community authorship. Contributors are encouraged to submit
128 new notebooks through pull requests, document their methods clearly, and generalise
129 workflows so that they remain useful to others. This is particularly valuable as it is
130 relatively rare for the computer code underlying scientific analyses to be peer reviewed
131 with the same care as the manuscript itself. The COSIMA community provides several
132 examples of the recipe ecosystem being extended into full research projects built on
133 peer-authored software repositories on Github, where analysis workflows, methods, and
134 implementation choices are exposed to community scrutiny and reuse. In that sense, the
135 Cookbook functions both as a learning module for end users and as a framework for
136 teaching reproducible scientific communication through notebook design and peer review.
137 The easy access to data and analysis provided by the COSIMA Cookbook showcases best
138 practices to the broader oceanographic community and has enabled rapid community
139 adoption of the ACCESS ocean and sea-ice model configurations internationally, facilitating
140 to date well over [100 peer-reviewed papers](#) and more than 20 PhD projects to completion
141 with several other underway.

142 Author Order

143 The first two authors contributed equally; authors from the third position onward are
144 listed alphabetically by surname.

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